

Diksha Rade

Bachelor of Technology
in Artificial Intelligence and Data Science
Vishwakarma Institute Of Technology, Pune

+91-9420316679
diksha.rade24@vit.edu
GitHub
LinkedIn

Summary

Artificial Intelligence & Data Science undergraduate with industry internship experience in Artificial Intelligence, Machine Learning, Full-Stack Development, Computer Vision, and Generative AI. Proficient in Python, Django, FastAPI, OpenCV, REST APIs, MongoDB, PostgreSQL, Git, HTML, CSS, and JavaScript, with proven experience in designing scalable AI-powered applications, backend services, and full-stack web solutions. Published researcher with strong problem-solving, analytical, and software engineering skills, seeking to contribute to innovative, data-driven technology products

Education

Year	Degree/Certificate	Institute	CGPA/Percentage
2024-2027	B.Tech AI&DS (Present 3rd Year)	Vishwakarma Institute Of Technology, Pune	9.60
2024	HSC(XII)	CHM College, Ulhasnagar	80.60%
2022	SSC(X)	SPA Secondary School, Ambernath	94.40%

Professional Experience

- **HCL -Technologies** *Sep 2025 - Present*
  NLP/LLM Intern Remote
  – Contributed to the development of a Retrieval-Augmented Generation (RAG) pipeline for extracting structured information from unstructured technical documents.
  – Designed and implemented JSON schema-based extraction workflows using LangChain and Groq API to improve document processing.
  – Assisted in integrating LLaMA-based models into the RAG pipeline and collaborated on backend API development using FastAPI.
- **Infosys Springboard** *Feb 2026 - April 2026*
  Full-Stack Developer Intern Remote
  – Engineered responsive user interfaces and server-side application logic using full-stack web development technologies to build scalable web applications.
  – Implemented REST APIs, database integration, authentication, and CRUD operations while ensuring application security and performance optimization.
  – Participated in Agile development, code reviews, debugging, testing, and Git-based version control to support end-to-end software delivery.

Projects

- **OrbitalGuard – AI-Powered Satellite Mission Planning & Collision Avoidance System**
  – Engineered an AI-driven decision support system that performs satellite launch planning, orbital slot recommendation, real-time space debris detection, and collision risk assessment using computer vision and trajectory prediction models.
  – Implemented automated collision avoidance by generating optimal orbital maneuver recommendations, reducing mission planning complexity through predictive analytics and orbital simulation.
- **SkillBridge – AI-Enabled Freelancer & Recruiter Collaboration Platform**
  – Developed a scalable full-stack recruitment platform supporting freelancer portfolios, intelligent job matching, secure authentication, interview scheduling, milestone-based project tracking, and escrow-enabled payment workflows.
  – Designed a role-based architecture with recruiter dashboards, task assignment, project lifecycle management, real-time communication, and analytics to streamline end-to-end hiring operations.
- **SignSpeak AI – Real-Time Speech-to-Sign Language Translation**
  – Developed an AI-powered accessibility application that converts live speech into real-time sign language gestures using speech recognition and deep learning-based sign generation.
  – Designed a low-latency translation pipeline to improve communication accessibility for hearing-impaired users with continuous speech processing and gesture rendering.
- **AnganTrack – AI-Driven Child Health & Anganwadi Management Platform**
  – Built an intelligent healthcare platform integrating AI-based malnutrition and pediatric disease detection from images, beneficiary health monitoring, stock inventory management, and real-time analytics dashboards.
  – Implemented role-based access for Anganwadi workers, AI-powered chatbot assistance, automated growth tracking, government supply monitoring, visit scheduling, and centralized reporting for data-driven decision making.

Technical Skills

- **Programming Languages**: C/C++, Python
- **Web Technologies**: HTML5, CSS3, JavaScript, Django, FastAPI, REST APIs
- **AI & Machine Learning:**: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, OpenCV, YOLOv8, CNN
- **Databases**: MySQL, MongoDB, PostgreSQL
- **Tools and Frameworks**: Git, GitHub, VS Code, Postman, Jupyter Notebook, Google Colab
- **Core Concepts**: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), DBMS, Operating Systems, Computer Networks

Publications

- **AI-Driven Framework for Space Debris Detection, Trajectory Prediction, and Satellite Collision Avoidance**
  (Published in) : *International Conference on Recent Advances in Intelligent and Sustainable Technologies (RAIST 2026), IIIT Surat, India.*
  Proposed an AI-powered framework for satellite mission planning integrating space debris detection, trajectory prediction, collision risk assessment, and intelligent orbital maneuver recommendation using Computer Vision and Machine Learning.

Achievements

- **Solved 200+ Data Structures & Algorithms (DSA) and Programming Problems** across coding platforms, demonstrating strong problem-solving abilities, algorithmic thinking, and proficiency in efficient coding techniques.
- **Qualified for the Final Round in 10+ National-Level Hackathons**, collaborating in cross-functional teams to design and develop innovative AI, Machine Learning, and Full-Stack software solutions under competitive environments.

Certifications

- **NVIDIA -** Applications of AI for Anomaly Detection
- **Infosys Springboard -** Artificial Intelligence Foundation Certification
- **Infosys Springboard -** Front End Web Developer Certification
- **Infosys Springboard -** Introduction to Natural Language Processing